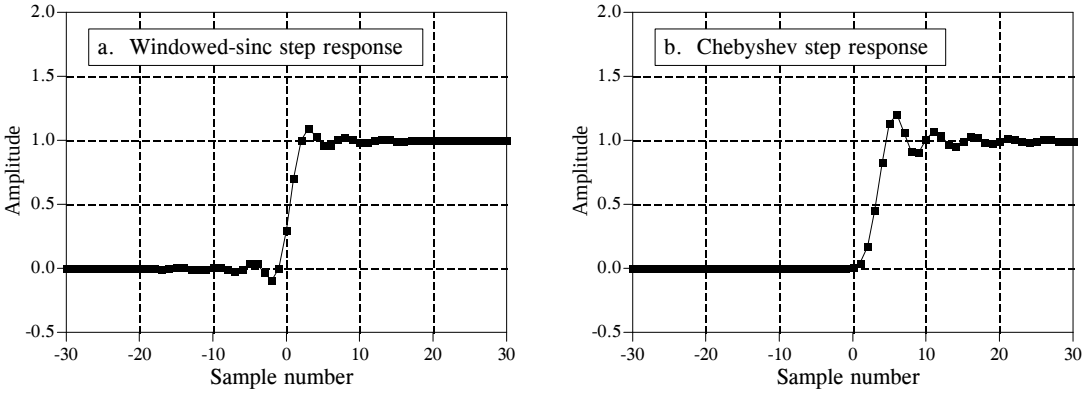

FIGURE 21-3
Windowed--sinc and Chebyshev step responses. The step responses are shown for a 51 point windowed-sinc filter and a 6 pole, 0.5% ripple Chebyshev recursive filter. Each of these filters has a cutoff frequency of 0.2. The windowed-sinc has a slightly better step response because it has less overshoot and a zero phase.

millivolt signal at 61 hertz that is riding on a 120 volt power line at 60 hertz. Figure 21-4 shows how these two filters compare when you need maximum performance. The recursive filter is a 6 pole Chebyshev with 0.5% ripple. This is the maximum number of poles that can be used at a 0.05 cutoff frequency with single precision. The windowed-sinc uses a 1001 point filter kernel, formed by convolving a 501 point windowed-sinc filter kernel with itself. As shown in Chapter 16, this provides greater stopband attenuation.

How do these two filters compare when maximum performance is needed? The windowed-sinc crushes the Chebyshev! Even if the recursive filter were improved (more poles, multistage implementation, double precision, etc.), it is still no match for the FIR performance. This is especially impressive when you consider that the windowed-sinc has only begun to fight. There are strong limits on the maximum performance that recursive filters can provide. The windowed-sinc, in contrast, can be pushed to incredible levels. This is, of course, provided you are willing to wait for the result. Which brings up the second critical issue: *speed*.

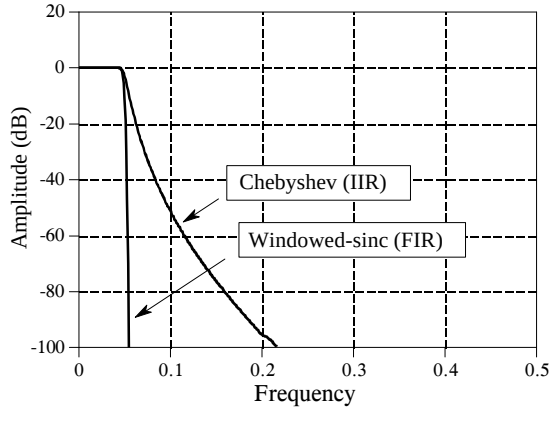

FIGURE 21-4
Maximum performance of FIR and IIR filters. The frequency response of the windowed-sinc can be virtually any shape needed, while the Chebyshev recursive filter is very limited. This graph compares the frequency response of a six pole Chebyshev recursive filter with a 1001 point windowed-sinc filter.